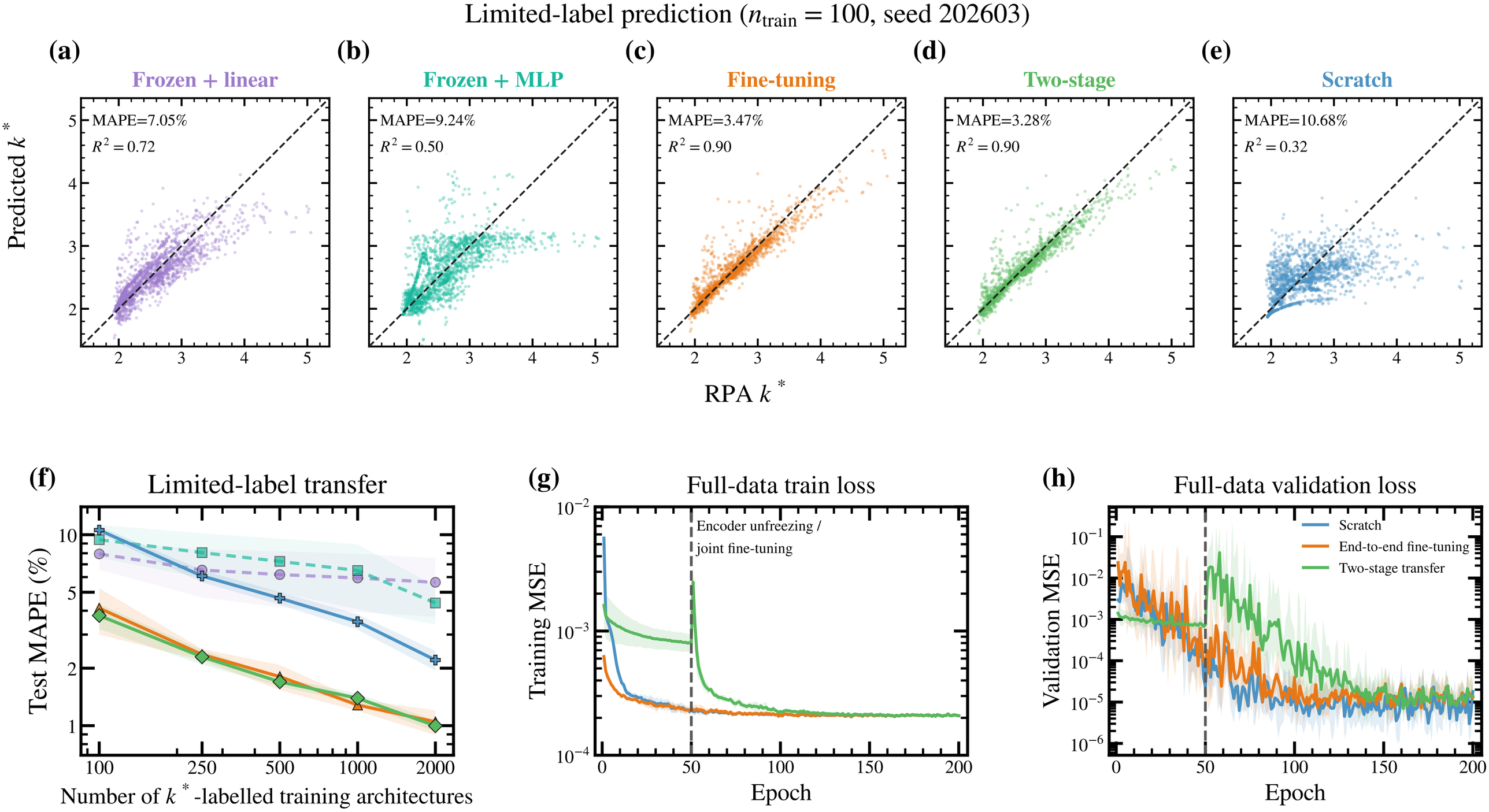

Limited-label prediction (n_{train} = 100, seed 202603)
(a)
Frozen + linear
MAPE=7.05%
R^2 = 0.72
(b)
Frozen + MLP
MAPE=9.24%
R^2 = 0.50
(c)
Fine-tuning
MAPE=3.47%
R^2 = 0.90
(d)
Two-stage
MAPE=3.28%
R^2 = 0.90
(e)
Scratch
MAPE=10.68%
R^2 = 0.32
Predicted k^*
RPA k^*
(f)
Limited-label transfer
Test MAPE (%)
Number of k^*-labelled training architectures
(g)
Full-data train loss
Encoder unfreezing / joint fine-tuning
Training MSE
Epoch
(h)
Full-data validation loss
Scratch
End-to-end fine-tuning
Two-stage transfer
Validation MSE
Epoch